Continuous random variables

Antonio de Leon

Probability and Statistics for Engineers Fall 2026

Monday, October 26	Densities, CDFs, and expectation
Wednesday, October 28	Exponential and normal models
Friday, October 30	Joint densities and marginals

Course support and deadlines

Ed Discussion: questions, announcements, and private messages. No Canvas Inbox.

Open Ed from Canvas All times Pacific Full contact details and meeting links

Instructor / TAs	Office hours	Optional, ungraded discussions
Antonio de Leon	Fri 3–4 p.m. (in person) GSC Room 204	—
Jeannine Romero	Tue / Thu 10–11 a.m.	01A: Fri 9:20–10:25 a.m. R Carson Acad 240 (in person)
Ehsan Mirafzali	Mon 3–5 p.m.	01B: Fri 10:40–11:45 a.m. (Zoom)
Seokjun Choi	Wed 3–4 p.m. 5:05–6:05 p.m.	01C: Wed 4–5:05 p.m. (Zoom)

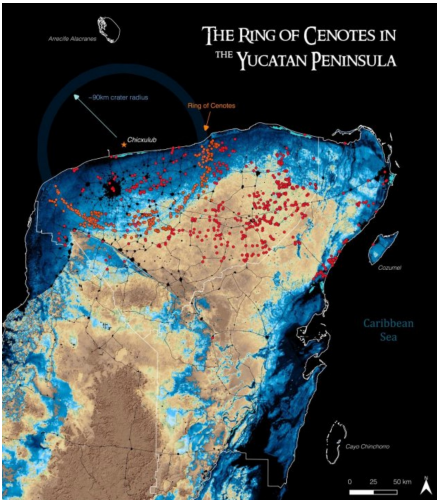
Upcoming deadlines HW and quizzes: **10 p.m. Pacific**

HW 3	**Tue, Oct 27**	Gradescope
Quiz 5	**Sun, Nov 1**	Canvas
HW 4	**Tue, Nov 3**	Gradescope

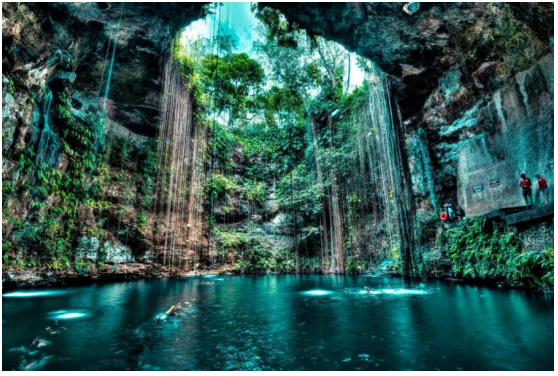
GSC: Graduate Student Commons. TA office-hour times link to Zoom.

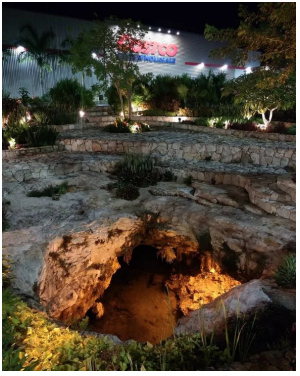
HW: 24-hour grace period. Quizzes: no grace period. Approved accommodations apply.




Arrecife Alacranes
THE RING OF CENOTES IN THE YUCATAN PENINSULA
~90km crater radius
Ring of Cenotes
Chicxulub
Cozumel
Caribbean Sea
Cayo Chinchorro
0 25 50 km

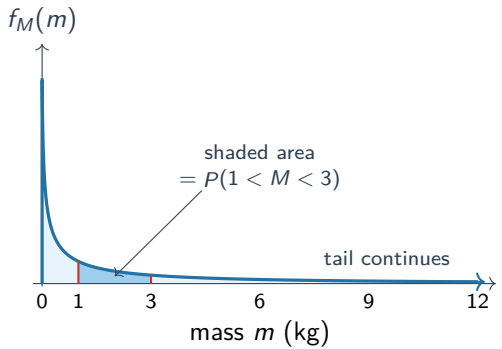






From a meteorite to a random mass

Let M be the reported recovered mass, in kg, of a recorded meteorite fall.



Mass is a measurement

The model allows any positive mass, including values between the labeled ticks.

Ask about an interval

What is the chance that M lies between 1 and 3 kg?

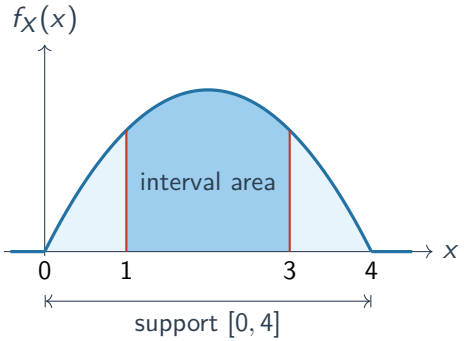
The shaded **area** gives that probability.

A probability density describes how probability is spread across possible values. The total area under the entire curve is 1.

Smooth approximation based on real recorded masses. Source: NASA / The Meteoritical Society, *Meteorite Landings*.

Reading a probability density

A **probability density function (PDF)** f_X describes probabilities by area. Now read a simpler sketch.



Read the axes
x: a value of X.
$f_X(x)$: the density height there.

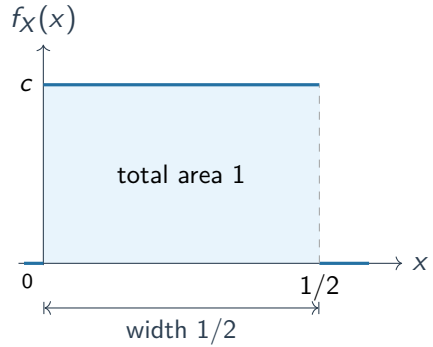
Identify the support
Here it is $[0, 4]$. The density is zero outside this range.

Read the shading
The darker area is $P(1 < X < 3)$.

Two requirements: every density height is nonnegative, and the **total area under the curve is one**.

A first density: where does the height come from?

To calculate an area exactly, model a waiting time X between 0 and $1/2$ minute with a **constant density** c, and zero density elsewhere.



All probability lies under this rectangle, so its area must equal 1.

$$\underbrace{\frac{1}{2}}_{\text{width}} \underbrace{c}_{\text{height}} = 1.$$

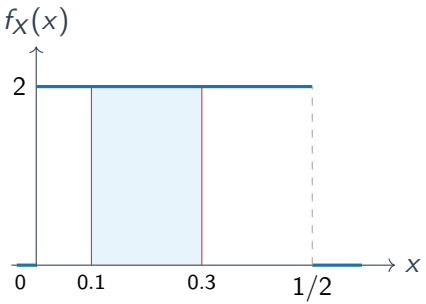
$$\boxed{c = \frac{1}{1/2} = 2.}$$

The height is 2 per minute. Multiplying by a width in minutes gives a probability.

A constant density on an interval is called a uniform distribution. Equal-length intervals inside it have equal probability.

Use area to find a probability

For the same model, $f_X(x) = 2$ for $0 < x < 1/2$, and 0 elsewhere.



Find $P(0.1 < X < 0.3)$

Width $= 0.3 - 0.1 = 0.2$.

Area $= 2(0.2) = 0.4$.

An integral gives the same area:

$$\int_{0.1}^{0.3} 2\,\mathrm{d}x = [2x]_{0.1}^{0.3} = 0.4.$$

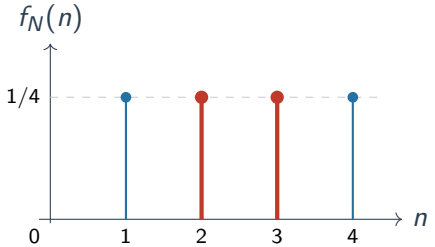
$$\boxed{P(a < X < b) = \int_a^b f_X(x)\,\mathrm{d}x} \qquad (a < b).$$

One point has zero width, so $P(X = c) = 0$. A density height, such as 2, is not a point probability.

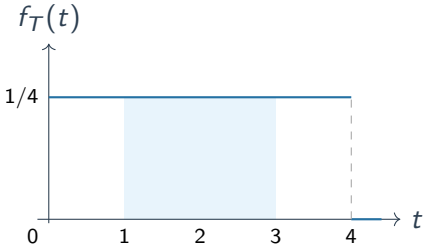
Including or excluding an interval's endpoints does not change its probability.

Compare with a discrete probability mass

Let N have four equally likely values $1, 2, 3, 4$. Let T have a uniform density on $(0, 4)$: width 4 requires height $1/4$.



Each marked height is a point probability.



The shaded area is an interval probability.

$$P(1 < N \leq 3) = \tfrac{1}{4} + \tfrac{1}{4} = \tfrac{1}{2}.$$

$$P(1 < T \leq 3) = \tfrac{1}{4}(3 - 1) = \tfrac{1}{2}.$$

PMF: add the point masses. PDF: find the area over the interval.

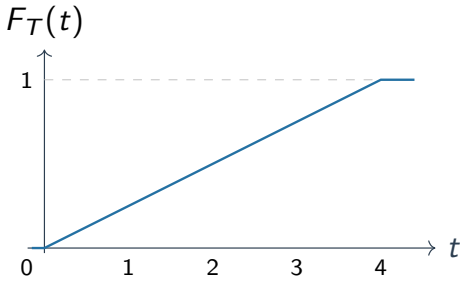
The symbol f is used for both functions. For a PDF, $f_X(x)$ is a height and $P(X = x) = 0$.

The CDF accumulates density area

$$F_X(t) = P(X \le t) = \int_{-\infty}^{t} f_X(u)\,\mathrm{d}u.$$

Continue with $T \sim \mathrm{Unif}(0, 4)$

$$F_T(t) = \begin{cases} 0, & t \le 0, \\ \displaystyle\int_0^t \frac{1}{4}\,\mathrm{d}u = \frac{t}{4}, & 0 < t < 4, \\ 1, & t \ge 4. \end{cases}$$



$$P(1 < T \le 3) = F_T(3) - F_T(1) = \tfrac{3}{4} - \tfrac{1}{4} = \tfrac{1}{2}.$$

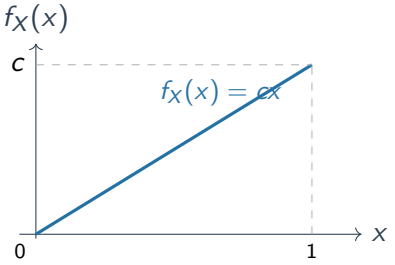
Fix the cutoff t. The integration variable u runs up to it.

A density with unequal weights

Now let the density increase: $f_X(x) = cx$ for $0 < x < 1$, and zero elsewhere. As with the rectangle, choose c so that total area is one.

$$\frac{1}{2}\underbrace{(1)}_{\text{base}}\underbrace{c}_{\text{height}} = 1, \quad c = 2.$$

Check by integration: $\int_0^1 cx\,\mathrm{d}x = 1$.



The triangle has base 1 and height c.

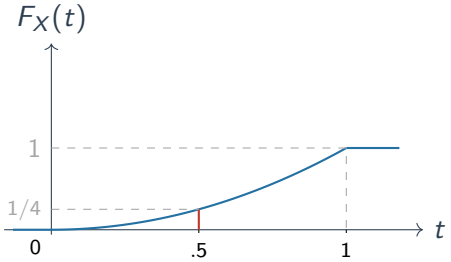
The increasing height gives more probability to equal-length intervals farther to the right.

The CDF of an increasing density

Let $f_X(x) = 2x$ for $0 < x < 1$, and zero elsewhere.

$$F_X(t) = \begin{cases} 0, & t \le 0, \\ \displaystyle\int_0^t 2x\,\mathrm{d}x = t^2, & 0 < t < 1, \\ 1, & t \ge 1. \end{cases}$$

Find $P(X > 1/2)$**.**



$F_X(1/2) = 1/4$ is the accumulated probability at the cutoff $1/2$.

The interior formula t^2 applies only for $0 < t < 1$.

Density from a CDF, and what its values mean

$F_X(x) = x^2$ for $0 < x < 1$. Here x names the cutoff. The CDF is 0 below the interval and 1 above it.

The CDF's slope gives the density

$$f_X(x) = \begin{cases} F_X'(x) = 2x, & 0 < x < 1, \\ 0, & \text{otherwise.} \end{cases}$$

The CDF is flat outside $(0, 1)$, so its derivative is zero there.

Evaluate three different quantities

Quantity	Value	Meaning
$f_X(3/4)$	$3/2$	density height
$F_X(3/4)$	$9/16$	probability $X \leq 3/4$
$P(X = 3/4)$	0	point probability

Integration accumulates density into a CDF. Differentiation recovers the density where the derivative exists.

At an isolated endpoint, the PDF value can be assigned arbitrarily without changing any probability.

Expectation as a weighted average

The idea of expectation stays the same: average the values using the distribution's weights.

Discrete X

$$\mathbb{E}[X] = \sum_{x \text{ in the support}} x\, f_X(x).$$

The PMF gives the probability weight for each possible value.

Continuous X

$$\boxed{\mathbb{E}[X] = \int_{-\infty}^{\infty} x\, f_X(x)\, \mathrm{d}x.}$$

The density supplies the weights throughout the range of values.

For a function of X, use the same distribution:

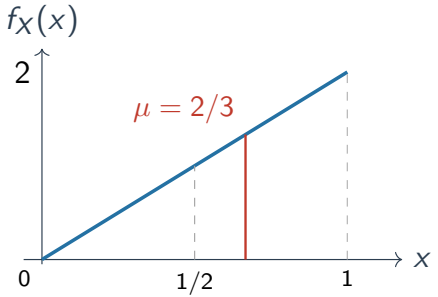
$$\mathbb{E}[g(X)] = \int_{-\infty}^{\infty} g(x) f_X(x)\, \mathrm{d}x.$$

Use $g(x) = x$ for the mean and $g(x) = x^2$ for the second moment.

The integration variable runs through the support. The result is a number. A finite expectation requires $\int |g(x)| f_X(x)\, \mathrm{d}x < \infty$.

Mean and variance of an increasing density

For $f_X(x) = 2x$ on $(0, 1)$, multiply the density by the quantity being averaged.



Larger values carry greater density weight, so the mean $2/3$ lies above $1/2$.

The variance formula from Week 4 still applies. Linearity and the affine scaling rules also carry over.

Uniform mean and variance

For $U \sim \text{Unif}(0,1)$, the density equals 1 on $(0,1)$.

$$\mathbb{E}[U] = \int_0^1 u \, \mathrm{d}u = \frac{1}{2}, \; \mathbb{E}[U^2] = \int_0^1 u^2 \, \mathrm{d}u = \frac{1}{3}.$$

$$\text{Var}(U) = \frac{1}{3} - \left(\frac{1}{2}\right)^2 = \frac{1}{12}.$$

More generally, a uniform distribution on (a, b) has

$$f_X(x) = \begin{cases} \dfrac{1}{b-a}, & a < x < b, \\ 0, & \text{otherwise}, \end{cases}$$

$$\mathbb{E}[X] = \frac{a+b}{2}, \; \text{Var}(X) = \frac{(b-a)^2}{12}.$$

The mean is the midpoint. A wider interval gives a larger variance.

Practice: a decreasing density

Suppose $f_W(w) = c(1 - w)$ for $0 < w < 1$, and zero elsewhere.

1. Find c and give $F_W(t)$ for every real t.
2. Compute $\mathbb{E}[W]$. Should it be below or above $1/2$?

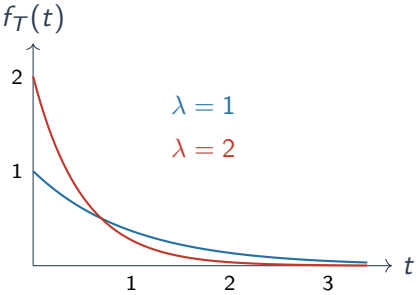
For review: HW4 Q1–3. Uniform formulas from today also support Q4(a).

An exponential waiting time

Assume a positive waiting time $T \sim \text{Exp}(\lambda)$, with rate $\lambda > 0$.

$$f_T(t) = \lambda e^{-\lambda t} I_{\{t>0\}}$$

Normalization



Time in minutes; rates per minute.

A larger rate places more probability near zero. Each curve has total area one.

The exponential law is a modeling assumption. Waiting times need not follow it.

CDF and survival function

As on Monday, integrate the density up to a fixed cutoff $t \geq 0$. Here u varies from 0 to t.

$$F_T(t) = \int_0^t \lambda e^{-\lambda u} \, \mathrm{d}u.$$

The survival function gives the probability that the wait exceeds t.
At $t = 0$, $F_T(0) = 0$ and $P(T > 0) = 1$.

Exponential moments and rate units

If $T \sim \text{Exp}(\lambda)$, then

$$\mathbb{E}[T] = \int_0^\infty t\lambda e^{-\lambda t}\,\mathrm{d}t = \frac{1}{\lambda},\ \mathbb{E}[T^2] = \frac{2}{\lambda^2}.$$

$$\text{Var}(T) = \frac{1}{\lambda^2},\ \text{SD}(T) = \frac{1}{\lambda}.$$

Example. A processing time has rate 2 per minute. Find its mean, variance, and $P(T > 0.5)$.

Rate has units of inverse time. The reciprocal rate is the mean waiting time.

The moment derivations by integration by parts appear in the optional appendix.

The remaining wait

Let $T \sim \text{Exp}(\lambda)$. For $s, t \geq 0$, suppose the wait has already exceeded s.



Derive $P(T > s + t \mid T > s)$ using event conditioning.

Under this model, elapsed time does not change the remaining waiting-time law.

The property is called **memorylessness**: t is an additional wait, while $s + t$ is a total wait.

A uniform wait has a different remaining wait

Let $U \sim \text{Unif}(0, 10)$, in minutes. The original wait is at most ten minutes. **Compare** $P(U > 7 \mid U > 4)$ with $P(U > 3)$.

Memorylessness is a property of the exponential model. A positive waiting time need not have that property.

The normal density

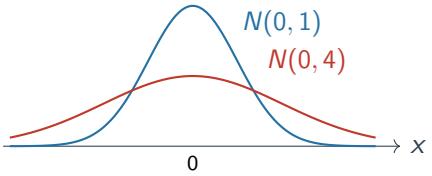
A normal model describes measurements that fluctuate symmetrically around a center. For $\sigma > 0$,

$$X \sim N(\mu, \sigma^2), \; f_X(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-(x-\mu)^2/(2\sigma^2)}, \; x \in \mathbb{R}.$$

The standard normal model

$$\phi(z) = \frac{1}{\sqrt{2\pi}} e^{-z^2/2}$$

$$\Phi(z) = P(Z \leq z) = \int_{-\infty}^{z} \phi(u)\,\mathrm{d}u$$



Same mean, different standard deviations.

Here $Z \sim N(0,1)$. The lower-case ϕ is a density and upper-case Φ is its CDF.

μ is the mean. σ^2 is the variance. Increasing σ spreads out the density while preserving total area one.

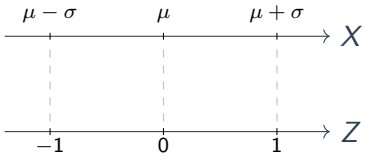
Standardization in a normal model

If $X \sim N(\mu, \sigma^2)$ with $\sigma > 0$, then

$$\boxed{Z = \frac{X - \mu}{\sigma} \sim N(0, 1).}$$

The event changes units.

$$X \leq b \iff Z \leq \frac{b - \mu}{\sigma}.$$



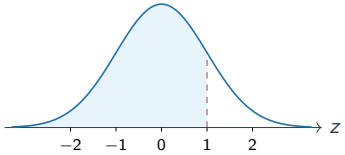
Subtract μ to center the scale. Divide by the SD σ to measure distance in standard deviations.

In $N(\mu, \sigma^2)$, the second parameter is the variance. Standardization uses its square root.

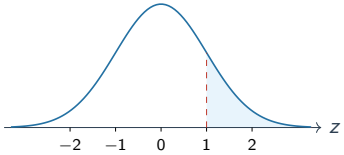
This gives a standard normal law because X is normal. Standardizing an arbitrary distribution does not make it normal.

Left, right, and interval probabilities

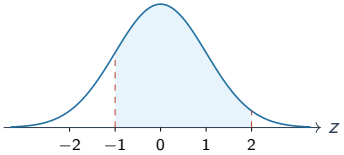
For $Z \sim N(0,1)$, $\Phi(z)$ is the area to the left of z.



$P(Z \le 1) = \Phi(1)$



$P(Z > 1) = 1 - \Phi(1)$



$P(-1 < Z < 2)$
$= \Phi(2) - \Phi(-1)$

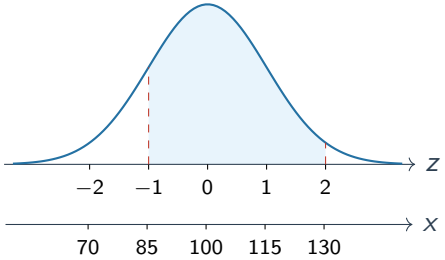
Use symmetry to express $\Phi(-z)$ through $\Phi(z)$.

z	0	1	2
$\Phi(z)$	0.500000000	0.841344746	0.977249868

The table gives numerical approximations. Shading extends into the unbounded tails.

A measurement between two limits

Assume a measurement $X \sim N(100, 225)$. Find $P(85 < X < 130)$.



First identify SD: $\sigma = \sqrt{225}$.

Original values align with standardized values.

Transform both endpoints. Centering and division by SD preserve their order.

Practice: normal tail probabilities

Assume $W \sim N(80, 16)$. Use $\Phi(1) = 0.841345$ and $\Phi(2) = 0.977250$.

1. Find $P(W > 84)$.
2. Find $P(|W - 80| > 8)$.

Parameters and probabilities

	$\text{Unif}(a, b)$	$\text{Exp}(\lambda)$	$N(\mu, \sigma^2)$
Range	(a, b)	$(0, \infty)$	$\mathbb{R}$
Mean	$(a + b)/2$	$1/\lambda$	μ
Variance	$(b - a)^2/12$	$1/\lambda^2$	σ^2
$F_X(t)$	$(t - a)/(b - a)$	$1 - e^{-\lambda t}$	$\Phi((t - \mu)/\sigma)$

λ is a rate. σ^2 is a variance. Standardization uses σ.

The uniform CDF formula applies for $a < t < b$ (outside: 0 or 1). The exponential formula applies for $t \geq 0$ (otherwise: 0).

Week 4 described a pair (X, Y) by the mass at each possible combination. We now use a joint density over the plane.

Discrete joint PMF

$$f_{X,Y}(x, y) = P(X = x, Y = y).$$

For an event region A, add the masses of the points in A.

Continuous joint PDF

$$P((X, Y) \in A) = \iint_A f_{X,Y}(x, y)\,\mathrm{d}x\,\mathrm{d}y.$$

Integrate the density over the part of A inside the support.

A joint density is nonnegative and has total integral one:

$$f_{X,Y}(x, y) \geq 0, \quad \iint_{\mathbb{R}^2} f_{X,Y}(x, y)\,\mathrm{d}x\,\mathrm{d}y = 1.$$

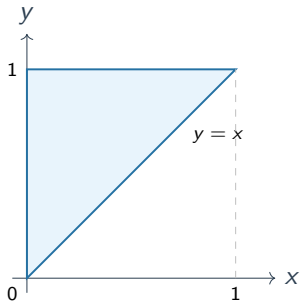
The joint PDF is zero outside its support. The region determines the limits, and the density supplies the probability weights.

The support of an ordered pair

Let X be a normalized start time and Y the finish time. Assume constant density on

$$S = \{(x, y) : 0 < x < y < 1\},$$

and zero elsewhere.



The inequalities describe the region together:

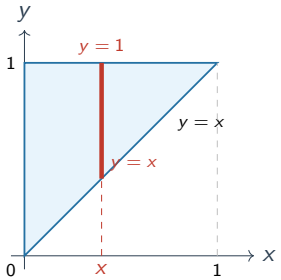
- $x > 0$: to the right of the vertical axis.
- $y > x$: above the diagonal $y = x$.
- $y < 1$: below the horizontal line $y = 1$.

The enclosing square contains pairs that the model never allows.

Uniformity over the triangle is the model assumption.

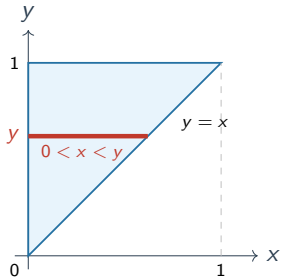
Reading the integration limits

The triangle $0 < x < y < 1$ can be described in either order.



Inner integral: fix x and vary y.

$$\int_0^1 \underbrace{\left[\int_x^1 f_{X,Y}(x,y)\,\mathrm{d}y\right]}_{\text{a function of } x} \mathrm{d}x$$



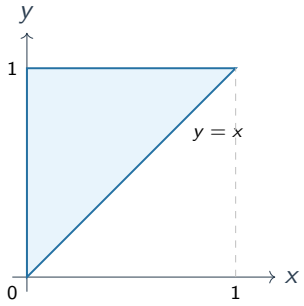
Inner integral: fix y and vary x.

$$\int_0^1 \underbrace{\left[\int_0^y f_{X,Y}(x,y)\,\mathrm{d}x\right]}_{\text{a function of } y} \mathrm{d}y$$

The inner integral returns a function. The outer integral then integrates that function.

A uniform law on a triangle

A normalized start/finish pair has constant joint density on $S = \{(x, y) : 0 < x < y < 1\}$.



$$f_{X,Y}(x, y) = c I_{\{0<x<y<1\}}$$

Use constant height $\times$ support area:

$$1 = c\left(\frac{1}{2}\right), \qquad c = 2.$$

Check with an iterated integral.

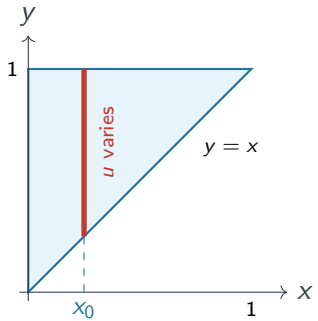
Uniformity over the triangle is the model assumption.

Uniformity over the triangle does not make either marginal distribution uniform.

A marginal density keeps one coordinate

To find $f_X(x_0)$, hold x_0 fixed and integrate over the Y coordinate.

$$f_X(x_0) = \int_{-\infty}^{\infty} f_{X,Y}(x_0, u)\,\mathrm{d}u.$$



Fixed: the argument x_0.
Varying: u, in the Y coordinate.

In the triangular model, set $x_0 = 1/4$:

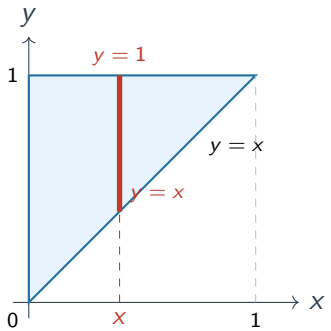
$$f_X(1/4) = \int_{1/4}^{1} 2\,\mathrm{d}u = \frac{3}{2}.$$

The result is a density height. It can exceed one.

Fixing the argument evaluates a density. It does not condition on an observation.

The marginal density of the start

For $f_{X,Y}(x,y) = 2I_{\{0<x<y<1\}}$, fix x and integrate over every compatible y.

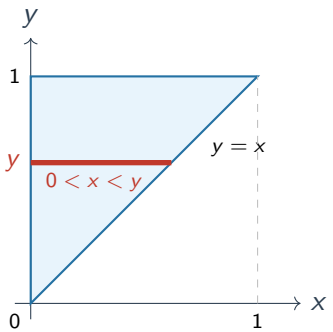


The projection onto the x-axis is $(0,1)$.

Find $f_X(x)$, including outside the projection.

The marginal density of the finish

Fix y. The compatible start coordinates satisfy $0 < x < y$.

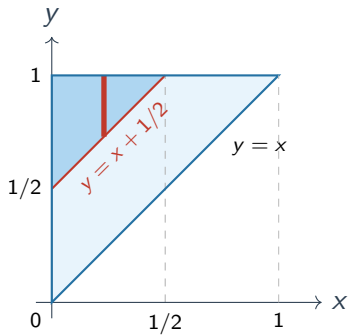


Find $f_Y(y)$ and check its integral.

The two marginals describe different coordinates of the same joint model.

A duration longer than one half

Find $P(Y - X > 1/2)$. Intersect the duration condition with $0 < x < y < 1$.

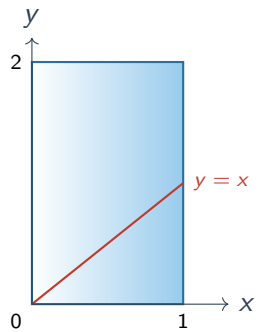


Read the limits from the shaded region.

Other order: $\int_{1/2}^{1} \int_{0}^{y-1/2} 2 \, \mathrm{d}x \, \mathrm{d}y$. Both integrals cover the same region.

A nonuniform joint density

Practice. Let $f_{X,Y}(x,y) = x I_{\{0<x<1,\ 0<y<2\}}$.



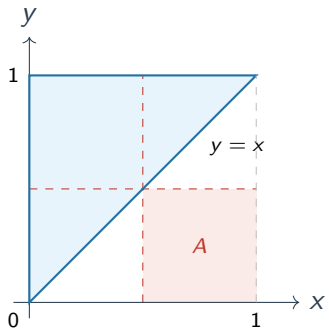
Density increases continuously with x.

The model is normalized: $\int_0^1 \int_0^2 x \, \mathrm{d}y \, \mathrm{d}x = 1$.

Shade $Y < X$, then calculate its probability.

Dependence from impossible combinations

Return to the triangular model. Consider $A = \{X > 1/2,\ Y < 1/2\}$.



Compare the joint-event probability with the product of marginal probabilities.

Independence would require these event probabilities to factor.

Independent continuous random variables

Independence means that, for every pair of sets A and B,

$$P(X \in A, Y \in B) = P(X \in A)\, P(Y \in B),$$

or equivalently $P(X \in A \mid Y \in B) = P(X \in A)$ whenever $P(Y \in B) > 0$. For variables with a joint density, independence holds exactly when a joint PDF can be written as

$$\boxed{f_{X,Y}(x, y) = f_X(x) f_Y(y)}.$$

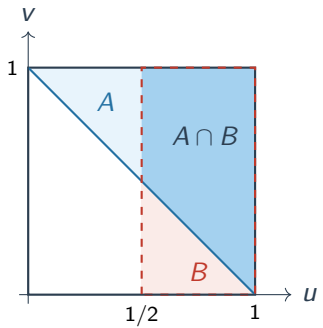
Example. Let U and V be independent $\mathrm{Unif}(0, 1)$ variables. Then

$$f_{U,V}(u, v) = \begin{cases} 1, & 0 < u < 1,\ 0 < v < 1, \\ 0, & \text{otherwise.} \end{cases}$$

Independence lets us build the joint density from the two marginal densities. The joint support must also agree.

An event condition within a joint model

Let U, V be independent Unif$(0, 1)$ variables. Define the events $A = \{U + V > 1\}$ and $B = \{U > 1/2\}$.



Find $P(A)$ and $P(A \mid B)$.

Dashed rectangle: B.

The sum uses U. Learning that $U > 1/2$ increases the chance that the sum exceeds one, even though U and V are independent.

Reading selections

Harchol-Balter, supplied 2024 version

Monday	pp. 134–137 and 141–142: density/CDF, uniform model, moments.
Wednesday	pp. 138–139, 142 (exponential); pp. 170–175 (normal moments and probabilities).
Friday	pp. 153–155: joint densities and marginals. Stop before conditional densities.

DeGroot–Schervish, fourth edition

Densities and moments	pp. 101–104, 111–112, 209–210; continuous expectation statement on pp. 213–214.
Named models	p. 303 and pp. 307–308 (normal); pp. 321–322, exponential portions only.
Joint laws	pp. 120–122 and the marginalization theorem on p. 132; p. 133 for the optional curved example.

Printed pages. Read selected passages alongside lecture; whole chapters are not assigned.

Exclude surrounding MGF, gamma, conditional-density, normal-sum, and CLT material. The appendix contains optional derivations and extensions.

Exponential mean and variance

Optional derivation or extension.

Use integration by parts on $[0, b]$, then let $b \to \infty$.

$$\mathbb{E}[T] = \int_0^\infty t\lambda e^{-\lambda t}\,\mathrm{d}t,\ \mathbb{E}[T^2] = \int_0^\infty t^2\lambda e^{-\lambda t}\,\mathrm{d}t.$$

The boundary terms vanish: $b^k e^{-\lambda b} \to 0$ for $k = 1, 2$.

Standard-normal moments

Optional derivation or extension.

Let Z have density ϕ. Its symmetry and decaying tails give $\mathbb{E}[Z] = 0$.

$$\phi'(z) = -z\phi(z),\ \mathbb{E}[Z^2] = \int_{-\infty}^{\infty} z^2\phi(z)\,\mathrm{d}z.$$

Integrate by parts using $z^2\phi(z) = -z\phi'(z)$.

Use finite bounds first. $z\phi(z) \to 0$ at both ends, and $\int |z|\phi(z)\,\mathrm{d}z < \infty$.

Why standardization works

Optional derivation or extension.

Let $X \sim N(\mu, \sigma^2)$ and set $Z = (X - \mu)/\sigma$, where $\sigma > 0$. **Find the CDF of Z.** Use $u = (x - \mu)/\sigma$ and $\mathrm{d}x = \sigma\,\mathrm{d}u$.

Thus $Z \sim N(0, 1)$. Since $X = \mu + \sigma Z$,

The CDF establishes the normal law. Mean zero and variance one alone do not.

Gaussian normalization

Optional derivation or extension.

Optional proof. Requires double integration and the polar area element.

Put $I = \int_{\mathbb{R}} e^{-z^2/2}\,\mathrm{d}z$. We seek the normalizing constant.

$$I^2 = \iint_{\mathbb{R}^2} e^{-(x^2+y^2)/2}\,\mathrm{d}x\,\mathrm{d}y.$$

Use $x = r\cos\theta$, $y = r\sin\theta$, and area element $r\,\mathrm{d}r\,\mathrm{d}\theta$.

$I < \infty$: for $|z| \geq 1$, $e^{-z^2/2} \leq e^{-|z|/2}$. Polar area is radius $\times$ radial width $\times$ angular width.

A normal percentile

Optional derivation or extension.

Optional application. A percentile is a threshold in a specified distribution.

For $X \sim N(\mu, \sigma^2)$ and $0 < p < 1$, find q_p such that $P(X \leq q_p) = p$.

For $N(100, 225)$, use $\Phi^{-1}(0.975) \approx 1.96$.

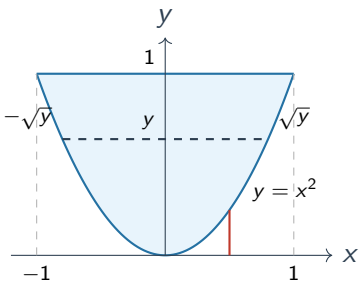
About 97.5% of this model's probability lies below the threshold.

A density on curved support

Optional derivation or extension.

Optional extension. The same slicing method applies to a curved boundary.

Let $f_{X,Y}(x,y) = cx^2 y I_{\{x^2<y<1\}}$.



Find c. The x-projection is $(-1, 1)$.

Marginals on curved support

Optional derivation or extension.

Optional extension. Here $f_{X,Y}(x,y) = (21/4)x^2 y I_{\{x^2<y<1\}}$.

Vertical slice

$-1 < x < 1$, $x^2 < y < 1$.

Horizontal slice

$0 < y < 1$, $-\sqrt{y} < x < \sqrt{y}$.

The horizontal slice uses both branches, $-\sqrt{y}$ and $\sqrt{y}$.